Import Data Using Transform Maps (Spreadsheet)

Phase 5 - Project Development

1. Development Overview

The **Project Development Phase** focuses on implementing the design created during the previous phases. In this project, the development process involves configuring ServiceNow to import data from a spreadsheet, store it in an Import Set Table, transform the data using a Transform Map, and finally create or update records in the target table.

The main development flow is:

Spreadsheet → Data Source → Import Set → Import Set Table → Transform Map → Target Table

The development process ensures that the requirements and design are converted into a working ServiceNow solution.

2. Development Objectives

The main objectives of the development phase are:

- Create the required ServiceNow configuration.
- Configure the spreadsheet as the source data.
- Create an Import Set Table.
- Configure the target table.
- Create a Transform Map.
- Map source fields to target fields.
- Implement data transformation logic.
- Configure Coalesce for duplicate handling.
- Implement validation and error handling.
- Test the complete data import process.
- Verify the final imported records.

3. Development Environment

The development will be performed in a ServiceNow development or test instance.

Required Components

- ServiceNow instance.
- Administrator/Developer access.
- Excel or CSV spreadsheet.
- Source data.

- Import Set.
- Import Set Table.
- Transform Map.
- Target table.
- Test data.

Development should be completed and tested in a non-production environment before deployment to production.

4. Spreadsheet Development

The first development activity is to prepare a standard spreadsheet template.

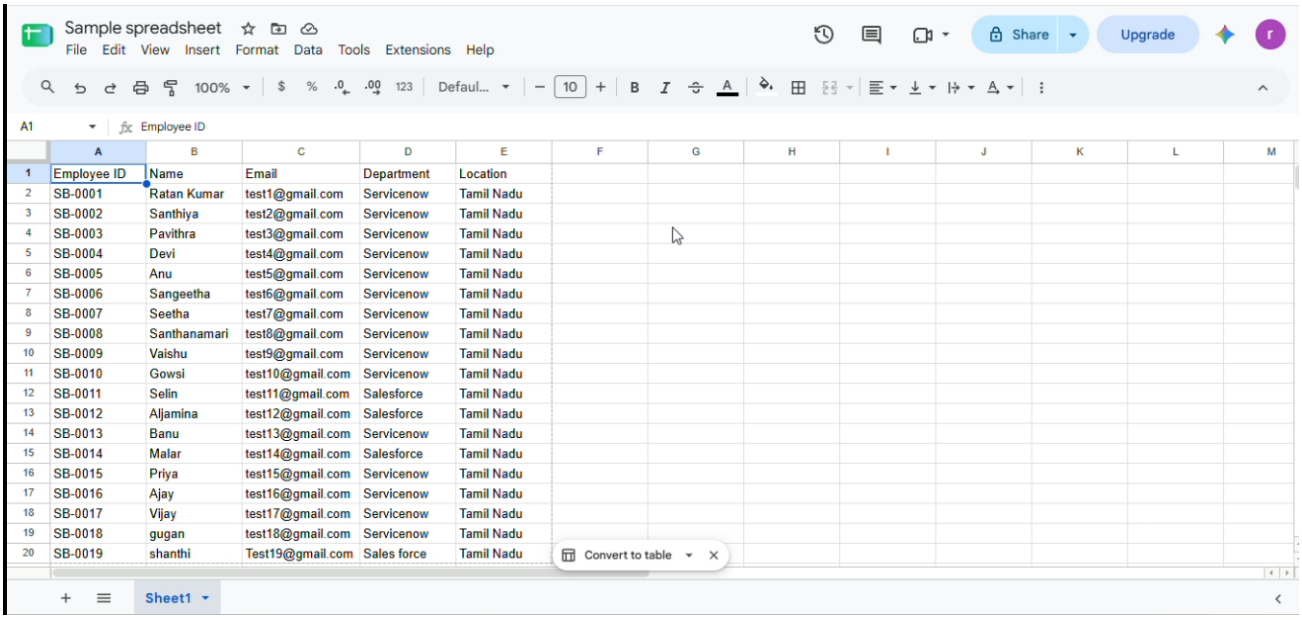
Example:

Employee ID	Employee Name	Email	Department	Joining Date
E001	Arun	arun@example.com	IT	10-01-2025
E002	Kumar	kumar@example.com	HR	15-02-2025
E003	Priya	priya@example.com	Finance	20-03-2025

Spreadsheet Requirements

The spreadsheet should:

- Have clear column headers.
- Contain valid data.
- Avoid unnecessary blank rows.
- Follow the expected date format.
- Contain unique Employee IDs.
- Include all mandatory fields.
- Avoid duplicate records where possible.



	A	B	C	D	E
1	Employee ID	Name	Email	Department	Location
2	SB-0001	Ratan Kumar	test1@gmail.com	Servicenow	Tamil Nadu
3	SB-0002	Santhiya	test2@gmail.com	Servicenow	Tamil Nadu
4	SB-0003	Pavithra	test3@gmail.com	Servicenow	Tamil Nadu
5	SB-0004	Devi	test4@gmail.com	Servicenow	Tamil Nadu
6	SB-0005	Anu	test5@gmail.com	Servicenow	Tamil Nadu
7	SB-0006	Sangeetha	test6@gmail.com	Servicenow	Tamil Nadu
8	SB-0007	Seetha	test7@gmail.com	Servicenow	Tamil Nadu
9	SB-0008	Santhanamari	test8@gmail.com	Servicenow	Tamil Nadu
10	SB-0009	Vaishu	test9@gmail.com	Servicenow	Tamil Nadu
11	SB-0010	Gowsi	test10@gmail.com	Servicenow	Tamil Nadu
12	SB-0011	Selin	test11@gmail.com	Salesforce	Tamil Nadu
13	SB-0012	Aljamina	test12@gmail.com	Salesforce	Tamil Nadu
14	SB-0013	Banu	test13@gmail.com	Servicenow	Tamil Nadu
15	SB-0014	Malar	test14@gmail.com	Salesforce	Tamil Nadu
16	SB-0015	Priya	test15@gmail.com	Servicenow	Tamil Nadu
17	SB-0016	Ajay	test16@gmail.com	Servicenow	Tamil Nadu
18	SB-0017	Vijay	test17@gmail.com	Servicenow	Tamil Nadu
19	SB-0018	gugan	test18@gmail.com	Servicenow	Tamil Nadu
20	SB-0019	shanthi	Test19@gmail.com	Sales force	Tamil Nadu

5. Data Source Development

A **Data Source** will be configured in ServiceNow to define the spreadsheet as the source of data.

The Data Source configuration identifies:

- Source type.
- File format.
- File location/upload method.
- Import Set Table.
- Import-related configuration.

The Data Source provides the connection between the spreadsheet and the ServiceNow Import Set process.

6. Import Set Table Development

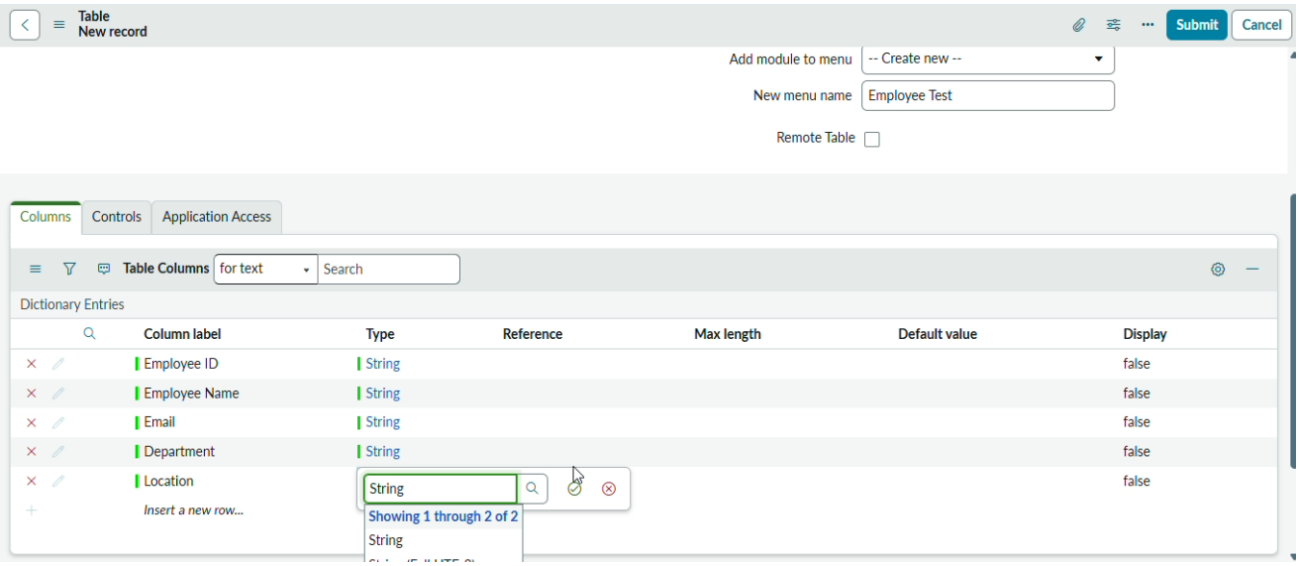
The next step is to create the **Import Set Table**.

The Import Set Table acts as a staging table where the spreadsheet data is temporarily stored before transformation.

Example fields:

Import Set Field	Type
u_employee_id	String
u_employee_name	String
u_email	String
u_department	String
u_joining_date	Date

The field structure should match the spreadsheet columns.



7. Target Table Development

The target table will store the final processed records.

For example:

Target Field	Type
Employee ID	String
Name	String
Email	Email/String
Department	Reference/String
Date of Joining	Date

The target table should be configured before creating the final Transform Map.

8. Transform Map Development

The **Transform Map** is the core component of the development.

It connects the Import Set Table with the target table.

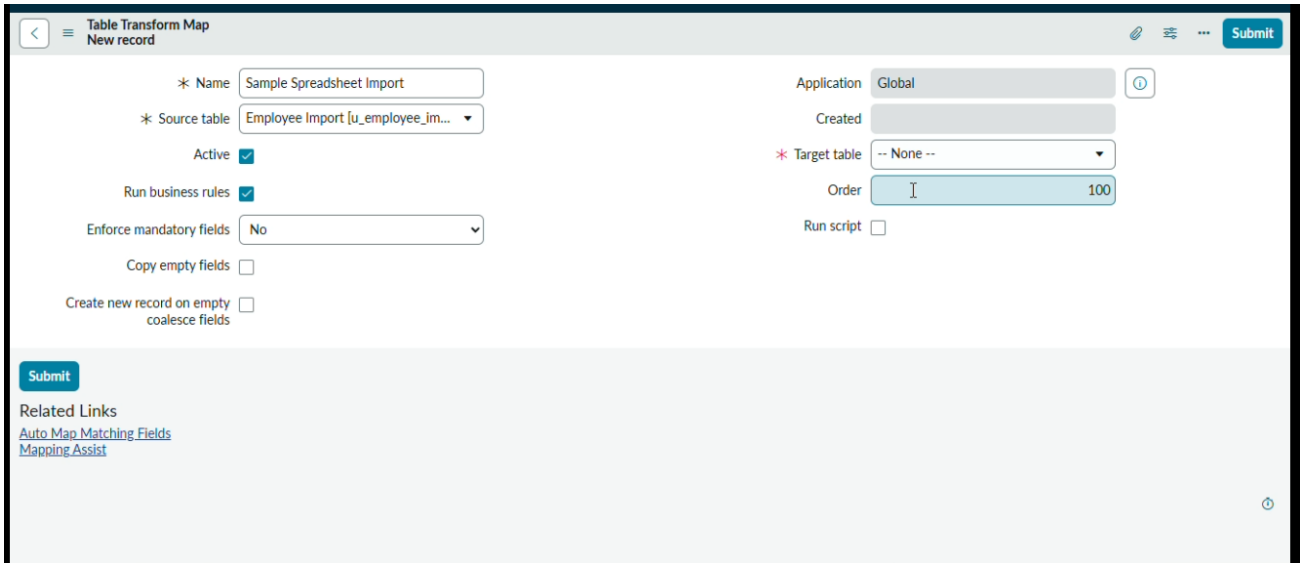
The Transform Map will define:

- Source table.
- Target table.
- Field mappings.
- Coalesce field.
- Transformation logic.
- Optional scripts.
- Error handling.

Example

```
Import Set Table                    Target Table

Employee ID        ------------->  Employee ID
Employee Name      ------------->  Name
Email              ------------->  Email
Department         ------------->  Department
Joining Date       ------------->  Date of Joining
```



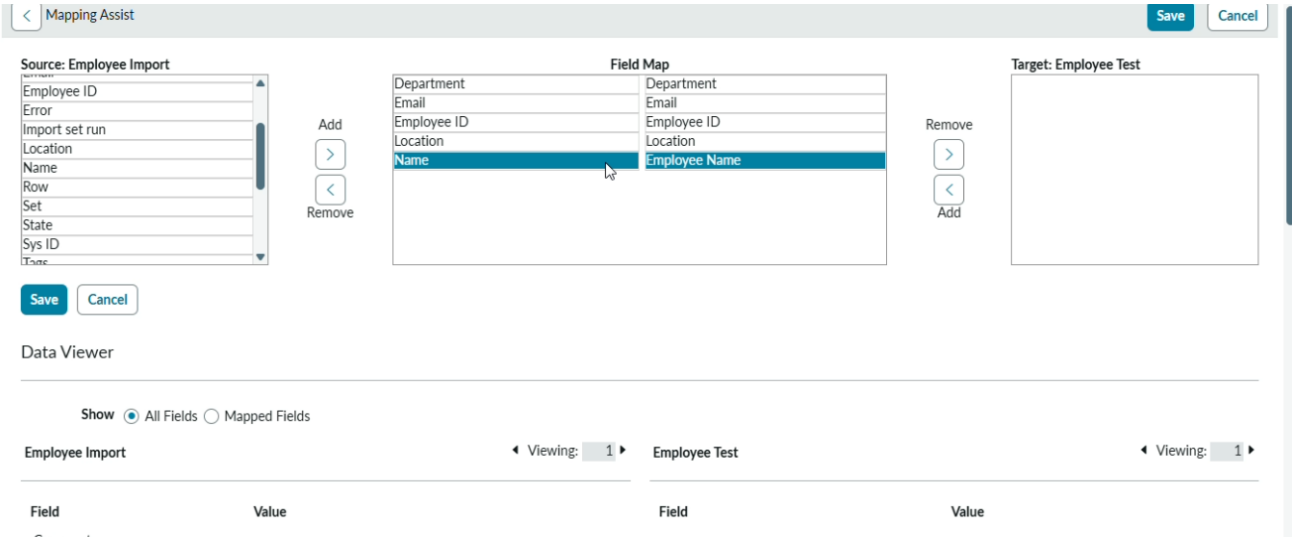
9. Field Mapping Development

Field mappings will be configured between source and target fields.

Example:

Source Field	Target Field
u_employee_id	Employee ID
u_employee_name	Name
u_email	Email
u_department	Department
u_joining_date	Date of Joining

Each mapping should be tested individually to ensure that the correct values reach the correct target fields.

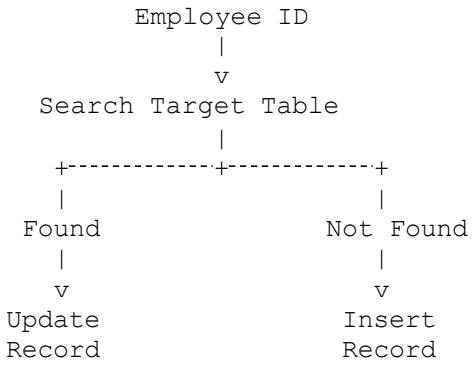


10. Coalesce Development

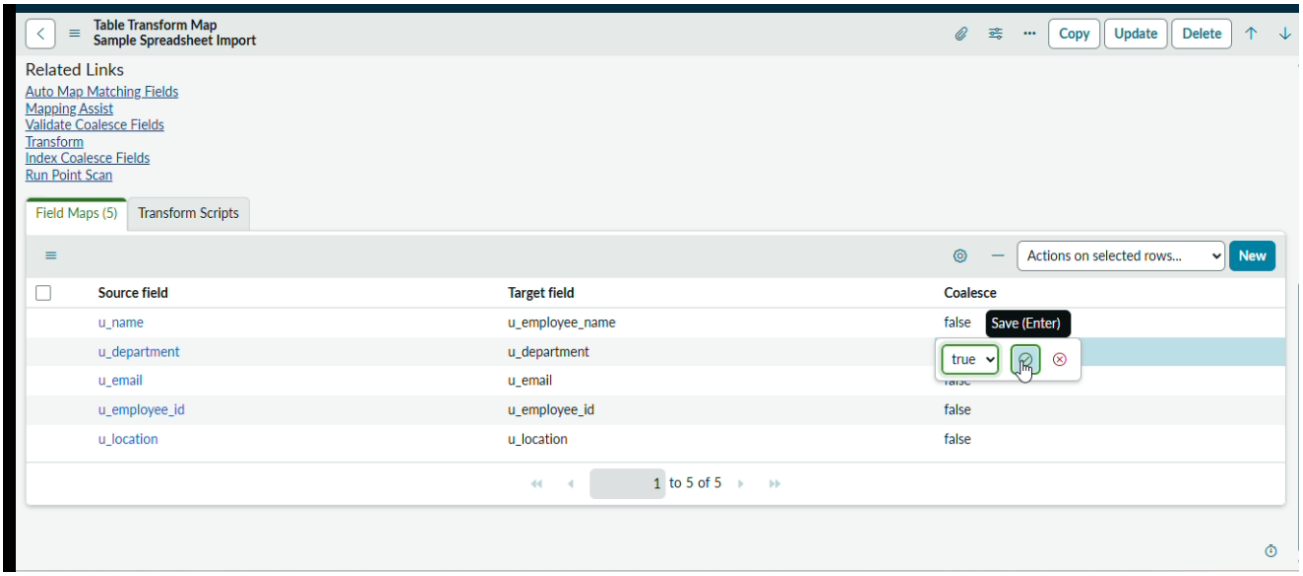
Coalesce will be configured to identify existing records.

For this project, **Employee ID** can be used as the unique identifier.

Development Logic



This ensures that the same employee does not unnecessarily receive multiple records.



11. Data Transformation Development

Transformation logic will be implemented wherever the source data requires modification.

Possible transformation activities include:

- Converting date formats.
- Removing unnecessary spaces.
- Standardizing text.
- Converting values to appropriate formats.
- Mapping source values to reference fields.
- Handling empty values.

For example, if department values in the spreadsheet are inconsistent, transformation logic can standardize them before inserting them into the target table.

12. Transform Script Development

If normal field mapping is not sufficient, Transform Scripts can be developed.

Transform Scripts can be used for:

- Complex data conversion.
- Conditional processing.
- Custom validation.
- Reference lookup.
- Data manipulation.
- Business-specific transformation logic.

For example:

```
If Department = "IT"
    Map to IT department
Else If Department = "HR"
    Map to HR department
Else
    Mark the record for review
```

Scripts should only be introduced when required because simple field mappings are easier to maintain.

13. Data Validation Development

Validation logic should be implemented to ensure data quality.

The development should verify:

- Employee ID is available.
- Employee Name is not empty.
- Email is valid.
- Department exists.
- Joining Date is valid.
- Required fields are populated.
- Duplicate records are handled.

Invalid data should be identified during the import/transform process.

14. Error Handling Development

Error handling will be implemented for failed records.

Possible errors include:

- Missing mandatory fields.
- Invalid email addresses.
- Incorrect date formats.
- Invalid department values.
- Mapping errors.
- Transformation failures.
- Duplicate or conflicting records.

The development should allow users to identify failed records, correct the source data, and re-import the corrected information.

15. Import Execution Development

Once all configurations are completed, the import process will be executed.

Execution Flow

```
1. Prepare Spreadsheet
          ↓
2. Upload Spreadsheet
          ↓
3. Load Data into Import Set
          ↓
4. Store Data in Import Set Table
          ↓
5. Execute Transform Map
          ↓
6. Validate Data
          ↓
7. Apply Transformation
          ↓
8. Check Coalesce
          ↓
9. Insert/Update Target Record
          ↓
10. Review Import Results
```

16. Testing During Development

Testing will be performed continuously during development rather than only at the end.

Test Case 1 – New Record

Input: New Employee ID

Expected Result: A new record should be created in the target table.

Test Case 2 – Existing Record

Input: Existing Employee ID

Expected Result: Existing record should be updated instead of creating a duplicate.

Test Case 3 – Missing Mandatory Field

Input: Employee Name is empty.

Expected Result: The record should be identified as invalid or failed.

Test Case 4 – Invalid Data

Input: Incorrect email or date format.

Expected Result: Appropriate validation/error handling should occur.

Test Case 5 – Multiple Records

Input: Spreadsheet containing multiple employee records.

Expected Result: All valid records should be processed successfully.

17. Development Verification

After implementation, the following areas should be verified:

Component	Verification
Spreadsheet	Correct format and data
Data Source	Correct source configuration
Import Set	Data successfully loaded
Import Set Table	Source data stored correctly
Transform Map	Transformation executes successfully
Field Mapping	Correct source-to-target mapping
Coalesce	Existing records identified
Target Table	Records inserted/updated correctly
Error Handling	Invalid records identified

18. Performance Consideration

The development should also consider the number of records being imported.

For larger spreadsheets:

- Avoid unnecessary transformation scripts.
- Use efficient mapping logic.
- Minimize unnecessary database operations.
- Test the import with realistic data volumes.
- Monitor transformation execution time.

This helps ensure that the solution performs efficiently.

19. Security Development

The development should follow ServiceNow security requirements.

The solution should ensure that:

- Only authorized users can perform imports.
- Users have appropriate access to the Import Set.
- Users have the required access to the target table.
- Sensitive information is handled securely.
- Existing access controls are respected.

20. Development Deliverables

The main deliverables from the development phase are:

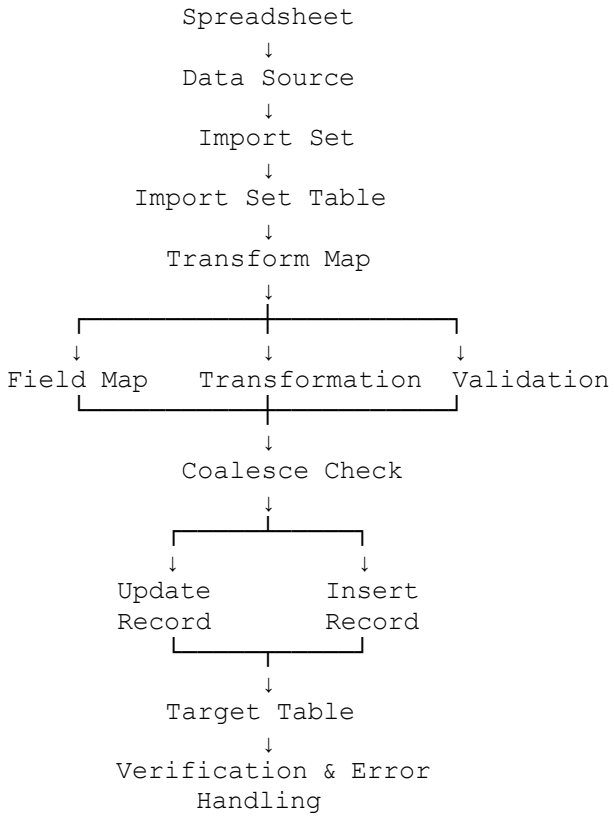
1. Standard spreadsheet template.
2. Data Source configuration.
3. Import Set configuration.
4. Import Set Table.
5. Target Table configuration.
6. Transform Map.
7. Field mappings.
8. Coalesce configuration.
9. Transformation logic.
10. Validation logic.
11. Error-handling configuration.
12. Test data.
13. Test results.
14. Development documentation.

21. Development Completion Criteria

The development phase will be considered complete when:

- The spreadsheet can be successfully uploaded.
- Data is loaded into the Import Set Table.
- Transform Map executes successfully.
- All required fields are mapped correctly.
- Data transformation works correctly.
- Coalesce prevents unnecessary duplicates.
- New records are inserted correctly.
- Existing records are updated correctly.
- Invalid records are identified.
- Errors can be reviewed and corrected.
- Test cases are successfully completed.

22. Final Development Flow



23. Final Project Development Statement

The Project Development Phase converts the planned design of the “Import Data Using Transform Maps (Spreadsheet)” project into a functional ServiceNow solution. During this phase, the spreadsheet, Data Source, Import Set Table, Transform Map, field mappings, Coalesce configuration, transformation logic, validation, and error handling are developed and integrated. The completed solution enables users to import spreadsheet data efficiently, transform it according to business requirements, prevent unnecessary duplicate records, and create or update records in the target ServiceNow table. The developed solution is then tested and verified to ensure that it meets the defined project requirements**.**